Lab 9: Inheritance Genetics — Genetics Problem Solving

BIOL-1

Name: ______________________________ **Date:** ______________________________

Learning Objectives

By the end of this lab, you will be able to:

1. Explain Mendel's two laws and connect them to meiosis.
2. Use Punnett squares to predict genotypic and phenotypic ratios for mono- and dihybrid crosses.
3. Solve problems involving incomplete dominance and codominance.
4. Determine blood type inheritance using the ABO allele system.
5. Predict outcomes of X-linked trait crosses.
6. Analyze a pedigree to determine the pattern of inheritance.
7. Explain polygenic traits and how environment affects phenotype.

Overview

Gregor Mendel discovered that traits are inherited in predictable patterns. By crossing pea plants and counting offspring, he uncovered the rules of genetics — rules that apply to all living things, including humans. In this lab, you will apply **Mendel's laws**, use **Punnett squares** to predict offspring, explore **blood types**, analyze **sex-linked traits**, interpret a **pedigree**, and examine the genetic basis of complex traits.

Materials

- This worksheet
- Scratch paper for additional Punnett squares

Part 1: Vocabulary and Mendel's Laws

Before solving genetics problems, you need the language. These are the essential terms for this module.

1. Fill in the definitions:

Term	Definition
Gene	
Allele	
Genotype	
Phenotype	
Homozygous	
Heterozygous	
Dominant	
Recessive	
Carrier	

2. Who is Gregor Mendel, and what organism did he study?

3. State Mendel's Law of Segregation in your own words. How does it connect to what happens in Meiosis I?

4. State Mendel's Law of Independent Assortment in your own words. What does it assume about genes on different chromosomes?

Part 2: Monohybrid Crosses

In pea plants, Tall (T) is dominant over Short (t). A heterozygous plant (Tt) looks tall but carries the short allele.

5. Cross a heterozygous tall plant (Tt) with a short plant (tt). Fill in the Punnett square:

	t	t
T		
t		

- Genotypic ratio:
- Phenotypic ratio:
- What percentage of offspring will be short?

6. Cross two heterozygous tall plants (Tt × Tt). Fill in the Punnett square:

	T	t
T		
t		

- Genotypic ratio:
- Phenotypic ratio:
- What is the chance of a homozygous recessive (tt) offspring?

7. What is a test cross, and what is it used for? What phenotypic ratio would you expect if the unknown parent is heterozygous?

Part 3: Dihybrid Crosses

When two traits are tracked at once, we use a 4×4 Punnett square. Mendel's Law of Independent Assortment tells us the two traits are inherited independently.

In pea plants: Tall (T) is dominant over short (t); Round seed (R) is dominant over wrinkled (r).

8. Cross two plants that are both heterozygous for two traits (TtRr × TtRr). What phenotypic ratio do you expect?

9. Out of 16 offspring expected in this cross, how many would be both Tall AND Round?

10. What law of Mendel does the dihybrid cross demonstrate? How does this law connect to what happens during Metaphase I of meiosis?

Part 4: Incomplete Dominance

In incomplete dominance, the heterozygote is a BLEND. Neither allele is fully dominant.

Example: In snapdragons, Red (RR) × White (R′R′) = Pink (RR′)

11. Cross two pink snapdragons (RR′ × RR′). Fill in the Punnett square:

	R	R′
R		
R′		

- Phenotypic ratio: ____ Red : ____ Pink : ____ White

12. How is incomplete dominance different from complete dominance? How is it different from codominance?

Part 5: ABO Blood Types — Codominance and Multiple Alleles

The ABO blood system demonstrates both codominance (I^A and I^B are codominant) and multiple alleles (three alleles: I^A, I^B, i). This system determines which antigens are on your red blood cells and is essential for safe blood transfusions.

13. Fill in the genotypes for each blood type:

Blood Type	Possible Genotype(s)	Antigens on Red Blood Cells
Type A		A antigen
Type B		B antigen
Type AB		Both A and B

Blood Type	Possible Genotype(s)	Antigens on Red Blood Cells
Type O		Neither

14. A mother is Type A (I^Ai) and a father is Type B (I^Bi). Fill in the Punnett square:

	I^B	i
I^A		
i		

15. What blood types are possible in their children?

16. Can they have a child with Type O blood? Explain using the Punnett square.

Part 6: Sex-Linked Traits

Color blindness is caused by a recessive allele on the X chromosome. Because males have only one X (XY), a single recessive allele is enough to cause the condition. Females (XX) need two copies.

Normal vision = X^B. Color blind = X^b.

17. A carrier mother (X^B X^b) and a normal-vision father (X^B Y) have children. Fill in the Punnett square:

	X^B	Y
X^B		
X^b		

18. What fraction of their sons will be colorblind?

19. What fraction of their daughters will be carriers?

20. Explain in your own words why X-linked recessive traits are more common in males.

Part 7: Pedigree Analysis

A pedigree is a family tree showing who in a family is affected by a genetic trait. Circles = females, Squares = males, Filled shapes = affected individuals.

Pedigree: Two unaffected parents (Generation I) have four children (Generation II): an affected son, an unaffected son, an unaffected carrier daughter, and an unaffected daughter. The carrier daughter later marries an unaffected man and has an affected son (Generation III).

21. Is this trait dominant or recessive? How do you know?

22. Is this trait autosomal or X-linked? What evidence supports your answer?

23. What are the genotypes of the Generation I parents?

24. What is a pedigree, and what can it tell you that a simple Punnett square cannot?

Part 8: Polygenic Traits and the Environment

Not all traits follow simple Mendelian patterns. Many human traits — like height, skin color, and eye color — are controlled by MANY genes working together.

25. What is a polygenic trait? Name two examples.

26. Mendel's pea plants were either tall OR short (two categories). Human height shows a continuous range from very short to very tall. Why is human height distributed as a bell curve instead of two distinct categories?

27. True or False: If you know someone's genotype, you always know their phenotype. Explain, giving a specific example where environment changes the phenotype.

28. A student says: "Genes determine everything about you — environment doesn't matter." Do you agree or disagree? Give a specific example to support your answer.